



optical-fibre cables.

Loose tubes are individually coloured for ease of identification, and individual fibres are colour-coded.

Creating test environments for optical-fibre cables

According to a market research report, the optical-fibre cable test equipment market can be classified based on the following:

- Type. Optical time-domain reflectometer, optical power meter, optical loss test set, remote fibre test system
- Form factor. Handheld, bench type
- Application. Installation, manufacturing, research
- End user. Telecom, cable TV, data centres

The total market is expected to reach US\$ 902.2 million by 2020 at an estimated CAGR of 4.86 per cent, from 2015 to 2020.

Optical-fibre cables are becoming an integral part of telecommunications. Need for maintenance and integrity of these networks are becoming increasingly important. Over 80 per cent of network failures happen due to improper cleaning of connectors at end-faces. An optical time-domain reflectometer can be used to measure the length of fault in single-mode or multi-mode optical-fibre cables.

Today, alarming changes are occurring in the design, installation and maintenance of optical-fibre cable networks, just as in the test and measurement (T&M) of optical-fibre cables.

An optical time-domain reflectometer cannot be used for short cables, as in LAN or building environment, as it is likely to show ghosts due to reflection at connectors. A light test set is easy to use and economical. An optical power meter is used to accurately measure the power of optical-fibre cable equipment or the power of signals passed through optical-fibre cables.

An optical-fibre cable network consists of such elements as fibres, connectors, splices, LEDs, laser sources, detectors and receivers. All of these need to be tested accurately for the efficient working of optical-fibre cable networks. The most important test is the insertion loss test of an installed optical-fibre cable network using a light

source performance meter or an optical loss test set. A visual fault locator uses a visible laser to detect faults. A red laser light is powerful enough for continuity checking, to trace it for several kilometres.

A remote fibre test system automates the process of detecting, locating and repairing faults on the outside of the fibre-optic plant. A remote fibre test system can monitor the fibre network 24 hours a day. The system automatically provides fast and precise notifications about any network problems. It detects and locates faults along the cable and sends clear alarm reports to the right personnel, at the right time.

Measuring losses in optical-fibre cables

There are two methods for measuring losses in optical-fibre cables, as follows:

- Single-ended loss method, which makes use of only the launch cable
 - Double-ended loss method, where the receiver cable is also attached to the test meter
- Various losses that occur in optical-fibre cables include:
- Attenuation loss, which is also known as signal loss or fibre loss; it is the loss of light energy as light pulses pass from one end to the other end of the cable
 - Light absorption loss, which is also known as material absorption loss
 - Scattering loss
 - Bend loss, which includes micro-bending and macro-bending losses
 - Optical-fibre cable joining loss

Devices in the market

“Key growth drivers of the T&M industry are introduction of new telecom technologies, explosion of data/video services, roll-out of 4G services, new quality of services (QoS), conformity to radiation specifications, spectrum management, etc,” opines Vishnu Goel, an optical-fibre cable industry expert.

Recently, Anritsu introduced access master MT 9085 series of handheld testers that provide accuracy and broad measurement capability during installation and maintenance of optical-fibre

cables in LTE and 5G backhaul, as well as metro and core networks. It incorporates a 20.3cm (8-inch) coloured touchscreen for enhanced operation, as well as integrates Anritsu’s fibre visualiser as a standard feature.

ED-4100 model by Euro Digital, available for ₹ 65,000, supports touch-sensitive LCD screen and multi-mode measurements.

Smart glass fibre OLTS-85/85 P by Viavi Solutions allows smart testing, and promotes fast and easy Tier 1 fibre testing based on TIA/ISO/IEC standards. It comes with dedicated multi-mode, single-mode and quad versions.

Challenges and improvement areas

An optical-fibre cable system is like a commodity. Challenges in T&M in the optical-fibre cable industry exist for both in-process and final quality testing, especially in standard telecom and datacom fibres. These challenges are mostly related to productivity, and are not technical in nature. This is due to availability of more-efficient and less-costly T&M equipment. Further, computer-based and Internet of Things (IoT) technologies give extremely precise and fast measurement possibilities.

It is true that test equipment purchased almost a decade ago are now obsolete. Many optical-fibre cable industries are now losing experienced test personnel due to newer trends. These days, it is easy to connect optical-fibre cables temporarily to test equipment. Automated test software and management information systems have further eased the T&M of optical-fibre cables.

“The IoT offers new possibilities for optical-fibre cable T&M industries,” says Henry Yaffe, founder and president of New Ridge Technologies. Through the IoT, it is possible to replace bulky T&M instruments with low-cost, ubiquitous, portable and smart devices. IoT-enabled optical spectrum analysers (such as NRT-8000 by New Ridge Technology Co.) are being greatly used by optical-fibre cable companies.

Due to the emergence of non-conventional, speciality optical-fibre cable manufacturers, high-quality unique measurement requirements exist. Optical-fibre cable manufacturers need more automated test equipment. **EFY**